

Mankiewicz extension on open connected domains

MathlibAnnex.IsometryEquiv.existsUnique_affineExtension

theorem

Glues local affine-isometry charts using their agreement on open overlaps.

Statement

Let E, F be real normed spaces, $S \subseteq E$ a nonempty connected open set, and $T \subseteq F$ an open set. If $f : S \rightarrow T$ is a bijective isometry, there is a unique surjective real affine isometry $A : E \rightarrow F$ satisfying $A(x) = f(x)$ for every $x \in S$.

Assumptions

The source is open, connected and nonempty; the target is open; and the specified map is a distance-preserving bijection. No completeness or finite dimensionality is required. Target connectedness is not an independent hypothesis.

Conclusion

Exactly one ambient affine isometry extends the entire given map f .

Proof route

Produce charts from matching small balls. Their agreement near an overlap point forces global equality, making the chart assignment locally constant and therefore constant on the connected source.

▼ Proof steps

1. At $x \in S$, openness gives positive radii R_s, R_t with $B(x, R_s) \subseteq S$ and $B(f(x), R_t) \subseteq T$. Set $R = \min(R_s, R_t) > 0$. Apply Construction of a local affine-isometry chart with this same f, x, R and the two smaller ball inclusions to obtain a surjective affine isometry A_x agreeing with f on $B(x, R/8)$.
2. The uniqueness mechanism is Agreement on a nonempty open set determines the affine map. If affine maps A, B agree on a nonempty open set u , choose $c \in u$ and $\varepsilon > 0$ with $B(c, \varepsilon) \subseteq u$. For $z \neq c$, set $a = \varepsilon / (2\|z - c\|) > 0$ and $y = c + a(z - c)$. Then $\|y - c\| = \varepsilon/2$, so $c, y \in u$. Affinity gives

$$0 = A(y) - B(y) = a(A(z) - B(z)),$$

since $A(c) = B(c)$. Cancel a ; at $z = c$ agreement is already known. Thus $A = B$ on the whole ambient space.

3. For each chosen chart A_x , let $r_x > 0$ be a radius of agreement. If $y \in S$ and $\|y - x\| < r_x$, then

$$u = (s \cap B(x, r_x)) \cap B(y, r_y)$$

is open, contains y , and is a region on which both A_x, A_y equal f . Step 2 gives $A_x = A_y$ globally. Thus $x \mapsto A_x$ is locally constant, exactly as required in Gluing locally agreeing affine isometries.

4. Choose $x_0 \in S$, using nonemptiness. On a connected space a locally constant map is constant: each fiber is open and its complement, a union of other open fibers, is open. Therefore $A_x = A_{x_0}$ for every $x \in S$. Put $A = A_{x_0}$; the local agreement at x gives $A(x) = A_x(x) = f(x)$. If another affine isometry has this property, it agrees with A on nonempty open S , so Step 2 proves uniqueness.

Main citations

- Exact theorem and proof
- Construction of a local affine-isometry chart
- Agreement on a nonempty open set determines the affine map
- Gluing locally agreeing affine isometries

Lean source signature (exact)

```
theorem existsUnique_affineExtension
  {s : Set E} {t : Set F} (f : s ≈i t)
  (hs : IsOpen s) (hsc : IsConnected s) (ht : IsOpen t) :
  ∃! A : E ≈a i[ℝ] F,
    ∀ x : s, A (x : E) = ((f x : t) : F)
```

In the source	Mathematical meaning
$(f : s \approx_i t)$	The given bijective isometry $f : S \rightarrow T$ between subsets of real normed spaces E, F .
$(hs : \text{IsOpen } s)$	The source S is open in E .
$(hsc : \text{IsConnected } s)$	The source S is connected and nonempty; the latter permits the choice of a base point.
$(ht : \text{IsOpen } t)$	The target T is open in F .
$\exists! A : E \approx^{\text{a i}}[\mathbb{R}] F$	There exists exactly one surjective real affine isometry $A : E \rightarrow F$ satisfying the following condition.
$\forall x : s, A (x : E) = ((f x : t) : F)$	The same A agrees with f at every source point $x \in S$, with both values viewed in F .

Exact source and proof.

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▼ **Exact content identity**

Declaration: `MathlibAnnex.IsometryEquiv.existsUnique_affineExtension`

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